

Infinite Traces in Strategies and Games

Ben Plummer, Corina Cîrstea

University of Southampton

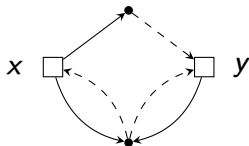
April, 2026

Two-player games

- ▶ Two players: controller ($\square$) versus environment ($\bullet$)

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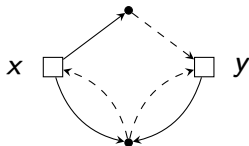
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- ▶ In synthesis: game arena + objective



Objective: “visit y infinitely often from x ”

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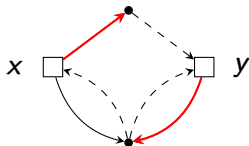


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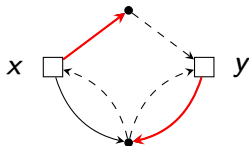


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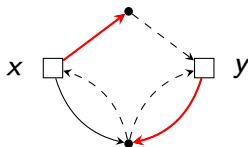


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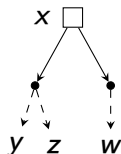
- ▶ Is there a winning strategy for $\square$?
 - ▶ There are infinitely many: strategies are history dependent
- ▶ Terminology:
 - ▶ *Strategy outcome*: the set of execution paths which conform to it
 - ▶ *Game outcome* at x : the set of strategies which start in x
 - ▶ Could be called the “strategy semantics”

... as a coalgebra

- ▶ We are concerned with the game arena
 - ▶ I.E. a *PP*-coalgebra $\delta : X \rightarrow PP(X)$
 - ▶ Each $U \in \delta(x)$ is controller move
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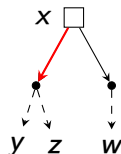
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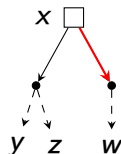
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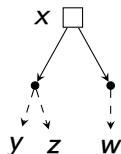
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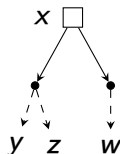


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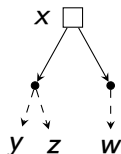


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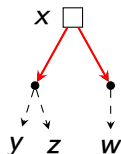


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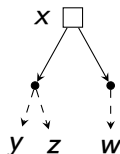


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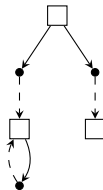
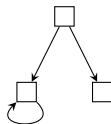


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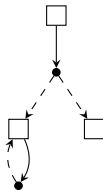
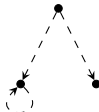
- ▶ Almost a monad structure on PP :
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 - ▶ There is a monad structure on $\mathcal{G}$

Strategies generalise executions paths and transition systems

$$P(X) \xrightarrow{P(\eta^P)} PP(X)$$

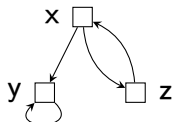


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Infinite executions (for transition systems)

- Given a transition system, compute the execution paths originating from each state



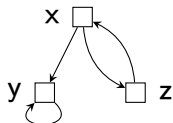
$$x \mapsto x(zx)^*(y^\omega + (zx)^\omega)$$

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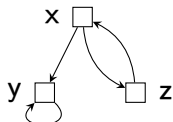
$$y \mapsto y^\omega$$

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- ▶ Work in **Rel**, the category of sets and relations
 - ▶ This moves from branching-time to linear-time semantics
 - ▶ $(-): \mathbf{Set} \rightarrow \mathbf{Rel}$ lifts functions into relations
 - ▶ Use $\mapsto$ to denote relations
 - ▶ **Rel** is **Pos**-enriched with subset inclusion

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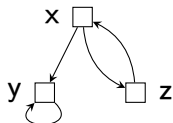
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 - ▶ **Rel** is **Pos**-enriched with subset inclusion
- ▶ Model a transition system $\delta: X \rightarrow X$ as an endorelation
- ▶ Let $F := X \times (-): \mathbf{Set} \rightarrow \mathbf{Set}$
 - ▶ The terminal F -coalgebra (Z, ζ) is $(X^\omega, \langle \text{hd}, \text{tail} \rangle)$
 - ▶ F lifts to $\bar{F}: \mathbf{Rel} \rightarrow \mathbf{Rel}$
 - ▶ Use $\gamma: X \rightarrow F(X)$ to record state information:
 - ▶ $\gamma(x) := \{(x, x') \mid x' \in \delta(x)\}$

Look at $\bar{F}$ -coalgebra morphisms into $(Z, \bar{\zeta})$

- Use $\gamma : X \rightarrowtail F(X)$ built from $\delta : X \rightarrowtail X$ from the previous slide

$$\begin{array}{ccc}
 X & \xrightarrow{\quad f \quad} & Z \ (\cong X^\omega) \\
 \gamma \downarrow & & \downarrow \bar{\zeta} \\
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- Is $(Z, \bar{\zeta})$ a terminal $\bar{F}$ -coalgebra?
 - Is f *uniquely determined* by the above property?

Non-canonical homomorphisms



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- ▶ The solution in [Jacobs, 2004], is to take the *largest homomorphism*
 - ▶ Yields $x \mapsto x \cdot X^\omega$, $y \mapsto y \cdot X^\omega$

This talk

We develop the following in a coalgebraic setting:

- ▶ Games
- ▶ Strategies
- ▶ Strategy outcomes
 - ▶ Limit-based definition
 - ▶ Largest homomorphism approach
- ▶ Game outcomes (the “strategy semantics”)
 - ▶ Limit-based definition

Defining a game

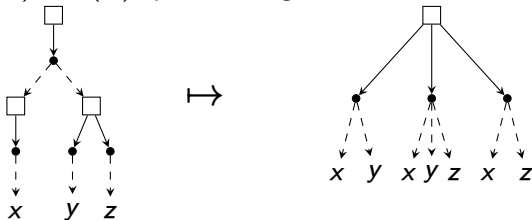
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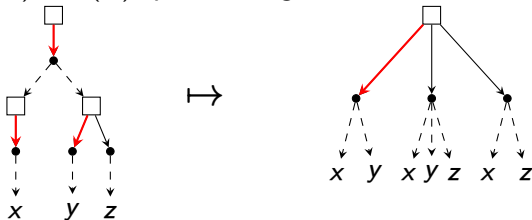
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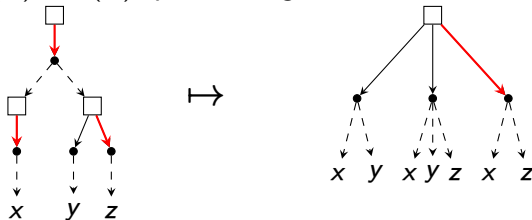
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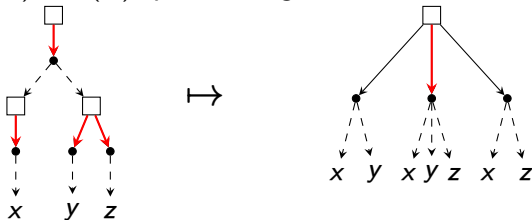
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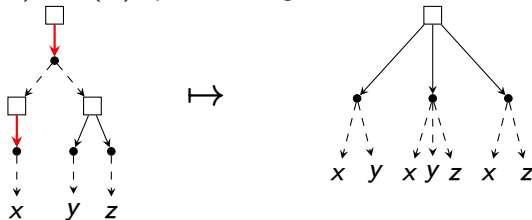
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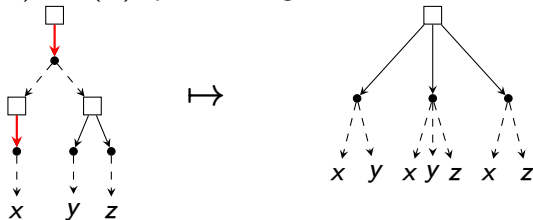
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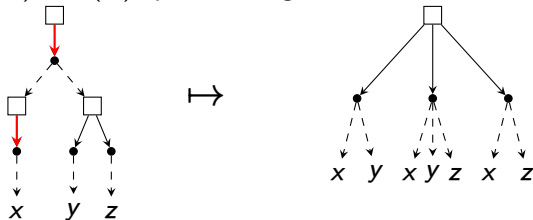
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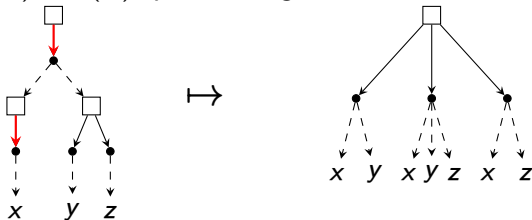


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- ▶ Use $\dashv\dashv$ to denote a **KI**($\mathcal{G}$) map
 - ▶ $\gamma : X \dashv\dashv F(X)$

Defining a strategy

- Consider a chain in **Rel**:

$$1 \xrightarrow{\sigma_*} X \xrightarrow{\sigma_0} X^2 \xrightarrow{\sigma_1} X^3 \xrightarrow{\sigma_2} \dots$$

- $\sigma_*(*) \subseteq X$ gives the starting states of the strategy

¹This can be stated categorically with a 2-cell in **KI**($\mathcal{G}$), which is also **Pos**-enriched

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¹This can be stated categorically with a 2-cell in $\mathbf{KI}(\mathcal{G})$, which is also **Pos**-enriched

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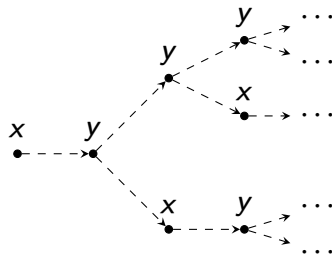
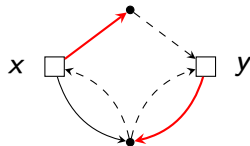
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- ▶ We can use $(-)^{\dagger} : \mathbf{Rel}_{\text{rt+sep}} \cong \mathbf{Set}$ to obtain a cochain $S_0 \leftarrow S_1 \leftarrow \dots$ in **Set**
 - ▶ Strategy outcome: take the limit of this cochain

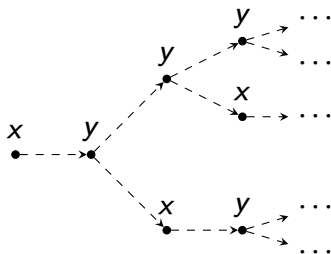
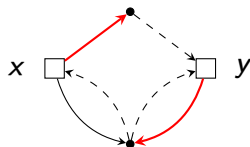
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Example



$$\{x\} \Rightarrow \{xy\} \Rightarrow \{xyx, xyy\} \Rightarrow \{xyxy, xyxx, xyyy\} \Rightarrow \dots$$

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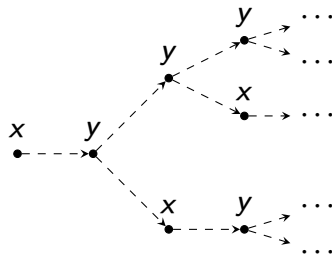
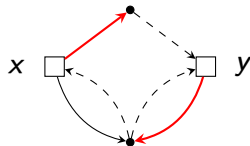
$$x \xrightarrow{\sigma_0} \{xy\}$$

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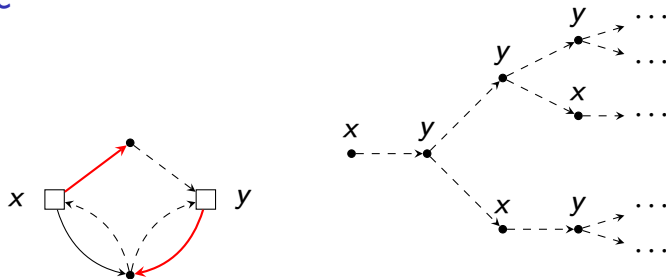
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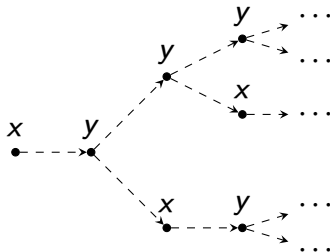
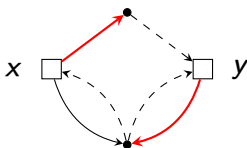


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where $\alpha := xyy^{*}$

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 \text{Pref}(\sigma) & \xrightarrow{h} & Z \quad (\cong X^\omega) \\
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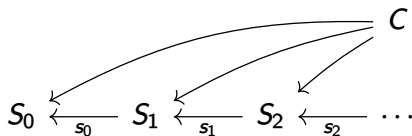
- ▶ $h(u)$ is the set of executions which complete u according to σ
 - ▶ Think of this as a greatest fixed point computation
 - ▶ Start with $h(u) = Z$, then refine backwards according to σ

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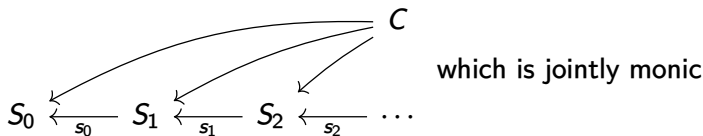


which is jointly monic

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- ▶ The greatest homomorphism is identified with the *largest jointly monic cone*:

- ▶ This is precisely the *limiting cone*
- ▶ Our two notions of outcome are the same

Game outcome as a limit

- ▶ Define $\gamma_n : X \dashv\dashv F^n(X)$ by iterating $\gamma : X \dashv\dashv F(X)$ in $\mathbf{KI}(\mathcal{G})$
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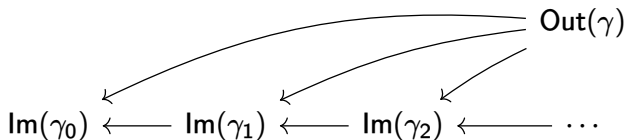
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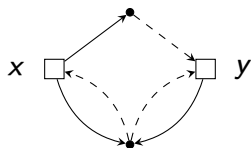
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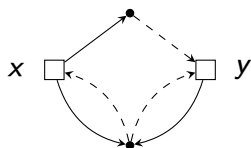
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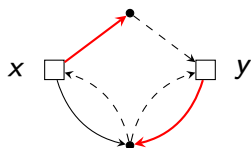
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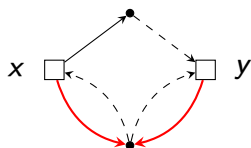
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- ▶ Future work:
 - ▶ Probabilistic opponents (MDPs, SSGs)
 - ▶ Weighted opponents (energy games)
 - ▶ Parity/Büchi games [Urabe et al., 2016]
 - ▶ Product construction of game with an automaton

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